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The role of urban transformation on inter-suburban commuting: evidence from Jakarta metropolitan area, Indonesia

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ABSTRACT

Post-suburbanization, shifting the focus of urban spatial development to its periphery, is becoming a global policy concern. Changing urban form affects travel behavior, where the traditional suburb-to-CBD commuting pattern has been contested by inter-suburban commuting. This empirical study investigates how employment decentralization and its associated heterogeneous suburban amenities have influenced commuting dynamics between suburbs in a developing country. Using the case of the Jakarta Metropolitan Area (JMA), Indonesia, spatial panel regression is applied to model inter-suburban commuting between 2008 and 2015. The results indicate that peripheral employment expansion due to industrial decentralization is followed by a significant increase in inter-suburban commuting. The mono-functional industrial subcenter has attracted more commuters from other suburbs. Yet, it also pushed some residents to commute away for jobs in service firms. The study has added a better understanding of urban form-travel behavior relationships that have seldom been researched from the Global South perspective.

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Post-suburbanization; commuting; spatial panel regression; Jakarta metropolitan area

Introduction

This article investigates how the changing suburban environment and employment deconcentration could influence inter-suburban commuting dynamics. The suburbs have been traditionally seen as developed areas outside the core, which mainly functioned as settlements (Harris, 2010). These urban fringe developments were linked to the CBD through a traditional commuting pattern. However, the initial suburbanization process had some breaks from the traditional development pattern (Wu & Phelps, 2008, p. 465). Due to employment decentralization, the metropolitan periphery turned into a multi-functional area with various activities containing industries and services while offering various employment (Soja, 2000). The process increased traditionally low suburban density through infill development, making suburban landscapes more similar to the urban core (Soja, 2011). In the suburbs, particular urban functions such as airports,

industrial and educational complexes, malls and commercial areas, theme parks, and amusement centers emerge next to residential areas. The new urban system has been termed “post-suburbia” (Borsdorf, 2004, p. 13; Kling et al., 1995), encompassing what existing literature coined as “edge city”, “edgeless city”, or “technoburb” (Wu & Phelps, 2008).

Post-suburbanization reflects changes in the suburban functions, making them attractive places, accompanied by the simultaneous decreasing importance of the traditional urban center (Phelps & Wu, 2011). While job concentration increases in the suburban landscape, post-suburbanization can be identified by several common features, such as a polycentric spatial pattern with mixed land-use and urban-rural blurring (Hudalah & Firman, 2012). The post-suburbia inherited many CBD characteristics. Beauregard (1995) argued that it becomes possible for suburbanites to do all their living, leisure, and recreation in an area like the edge city. Local mobility dominates commuting patterns in these areas, and most residents are isolated from the larger urban social politics. As a result, the overall urban commuting length is argued to decrease.

Contending perspectives argue that post-suburbanization makes commuting “no longer a one-way commuter stream towards the city center” (Borsdorf, 2004, p. 11), and most of the commuting are between suburbs (Richardson & Gordon, 2004). Consequently, personal mobility among suburbs and new logistic requirements for the decentralized service and leisure industries will increase commuting complexities (Borsdorf, 2004; Firman, 2011). These views imply that growing employment suburbanization does not necessarily reduce the need to commute and may actually increase the overall urban commuting length. Previous studies have supported the decreasing commuting length hypothesis, however, determined by various factors such as job-housing balance, the degree of sprawling, and workers’ socio-economic characteristics (Horner, 2004; Horner & Mefford, 2007; Sultana, 2000, 2002; Sultana & Weber, 2007, 2014; Weber & Sultana, 2008). Decreasing commuting is experienced by the inner-city, but new subcenters run into the opposite (Jun, 2020).

Current studies of commuting in the post-suburban era have mainly been discussed within the employment decentralization process, rooted in the western world tradition. These studies focus on service and retail-based decentralization, with less explanation about inter-suburban commuting (Bontje, 2004; Helbich & Leitner, 2009; Lucy & Phillips, 1997; Tzaninis & Boterman, 2018). Most studies on inter-suburban commuting have also been similarly conducted in developed countries. Studies have revealed that inter-suburban commuting occurs because the spatial transformation is caused not by worker residential adjustment and, instead, by increasing jobs-housing imbalances between the suburbs (i.e. Aguilera, 2005; Buchanan et al., 2006; Bontje, 2007; Gutiérrez & García-Palomares, 2007; Small & Verhoef, 2007, p. 5; Baum-Snow, 2010).

However, there are some essential differences between post-suburbanization processes in the global north vs global south. Respectively, theories developed from global north case studies have explanatory limitations when applied to the south. That is because certain issues that are important in the north are less so in the global south, resulting in a different post-suburban commuting pattern. Cities and regions in the global south have been experiencing premature deindustrialization, where growing jobs are dominated by small-scale industries as specialized parts of the wider global value chains (Pike, 2022, p. 8). The social transformation of employment and productivity culture

from agriculture to industry is stalled and remains incomplete (Levy, 2017). As a result, the jobs formed in these areas are typically occupied by less-skilled workers with a temporary arrangement. Thus, the typical residential choice for the working class in the global south is based on social exclusion and spatial segregation.

Similar to other developing countries, the common source of urban spatial segregation in Indonesia, like race, is negligible. Urban population distribution in Indonesia is more associated with poverty and informality (Rukmana et al., 2018). Most peripheral developments in JMA are inhabited by the lower-income population, unlike the middle-class suburbs in the US (Firman, 2011, p. 207). The capital influx and industrialization occur within the extended metropolitan region's interstitial areas, producing mixed uses of high-density residence, industry, and agriculture. The mixture of residential-agricultural-manufacturing suburbanization-based land-uses in these areas does not resemble that of the inner city and is not observed in western countries (Kurnia et al., 2022).

Such morphology, termed *desakota* (McGee, 1991), provides a background for Indonesia's post-suburbanization process. Most households in the *desakota* region tend to be locally employed, while some commute to the nearby city. The growing post-suburban area has become another informal job attraction for these households, while many formal suburban jobs are occupied by workers from the traditional core. The resulting reverse commuting and mixed settlement pattern reflects an early stage of the process and partially differentiates post-suburbanization in Indonesia from western countries (Firman & Fahmi, 2017).

In addition to the contextual background, previous studies on suburban commuting have paid little attention to the spatio-temporal aspect. As spatial units, suburbs have space-specific time-invariant variables. Although jobs and housing are generally balanced at the metropolitan level, some parts of the urban areas will have plentiful jobs and less housing, and vice versa (Horner, 2004). In the US, the distribution of office space across regions was uneven, whereas some metropolises have an edge city and others have multiple peripheral nodes (Lang et al., 2009). Edge city development has also produced varied settlement shapes and morphology typology across major European metropolises (Bontje & Burdack, 2005). Thus, different development trajectories created employment and population growth disparity among the suburbs resulting in the spatial distribution of various suburban attractions. The contrast generates inter-suburban commuting flows, which might differ from traditional suburb-to-core commuting. Failure to account for these variables may increase the risk of estimation bias (Elhorst, 2010).

Given these shortcomings, this paper provides an empirical analysis of the impact of metropolitan transformation on commuting dynamics in metropolitan suburbs. Using the Jakarta Metropolitan Area (JMA), Indonesia, as a case study, the potential spatio-temporal drivers of traditional and inter-suburban commuting flows are examined. The case of the JMA illustrates the complex relationship between urban form and mobility in one of the world's largest metropolises and contributes to the current studies in two ways. First, it improves the understanding of the linkage between post-suburbanization and changing commuting patterns in the suburbs of the Global South. While studies on employment decentralization in this region are still limited, they represent fundamental differences to the Global North regarding the lack of its residents' options in choosing where to live (Harris, 2010). Secondly, it provides empirical evidence of what Bontje

(2004) addressed as the impacts of the new employment subcenters in the metropolitan periphery on sustainable regional development.

This paper is organized into seven parts. Following the introduction, the second section provides the literature review of the nexus between urban form and commuting, the emergence of inter-suburban commuting, and the framework built upon these discussions. The third section highlights the urban development and mobility in the JMA. The fourth section presents the data and methods used in the analysis. The fifth section provides the results of descriptive and statistical modeling, followed by a discussion section. The last section summarizes the findings and offers a conclusion and avenues for future studies.

Urban form and the emergence of inter-suburban commuting

The urban form and commuting nexus

The modern urban reality was marked by rapid decentralization of economic activities, increased individual mobility, and a more fragmented spatial distribution of activities that led to the dispersed urban realms of multiple urban centers. In the suburbs, the transformation produced a land-use that significantly different from the traditional dormitory functions. The transformation process, however, varies across regions. In the US, suburban land-use change is mainly driven by massive office space development, exceeding the residential function and creating the so-called “edge city” (Beauregard, 1995; Garreau, 1991). The new suburban employment poles in Europe have more specific functions and lack independence from the traditional center. Having a less urban function, it was more of a “city edge” than an “edge city” (Bontje & Burdack, 2005). In China, the emerging edge cities arose from the government’s planned industrial parks in specific suburban locations (Zheng et al., 2017). The similarity of these examples came from their typical outer peripheral area and mixed land-uses.

The new urban form attracts much interest regarding mobility and accessibility issues (Clark & Kuijpers-Linde, 1994), since understanding the linkages between urban land development and commuting helps achieve a more sustainable city (Momeni & Antipova, 2022). Horner (2004) classified the land-use’s impact on commuting under three main topics: job-housing balance, excess commuting, and accessibility. In terms of balance, the polycentric urban form is seen as more efficient because it will reduce commuting times and costs (Clark & Kuijpers-Linde, 1994). The emergence of an additional employment subcenter is believed to juxtapose the workers to their job. Consequently, co-location between workers and their employment location will increase the job-housing balance.

Aguilera (2005) disproves those arguments by showing that commuting length in polycentric metropolises could also increase. Often, the majority of employment in the subcenters is occupied by the non-residents who live further from their place of work, and co-location between residential areas and the (sub)center is only experienced by a minority of inhabitants. The jobs-housing imbalance due to racial segregation, income differences, and workers’ ability to relocate their housing worsened the commuting length (i.e. Horner & Mefford, 2007; Sultana, 2002; Weber & Sultana, 2008). Greater gaps between subcenters will generate more excess commuting for the suburbanites.

Another aspect of land-use that influences commuting is accessibility. From a geographical aggregation point of view, the accessibility study covers the role of location, proximity, and potential interaction with other places (Horner, 2004). For instance, distance to the CBD has been used in many studies to explain commuting length variability. A consistent negative relationship between the distance to CBD and commuting length implies that widespread urban expansion will make commuters travel farther (Hu & Wang, 2016). The effect of secondary urban centers' existence relies on their size and proximity to workers. Longer commuting was generated by more prominent employment clusters that attract more workers and less accessible clusters (Sultana, 2000). The role of location is strongly related to the unequal distribution of resources across geographical spaces that create travel's supply and demand.

Other studies on accessibility focus on the individual and their activities, given its benefit and constraints. Assuming that workers seek utility maximization, there must be an adjustment of wages and rents level to prevent life quality from degradation and avoid an incentive for some workers to move (Roback, 1982). In their hypothetical model, Koster and Rouwendal (2013) prescribed a bid rent as a function of commuting costs between residential and business areas. The impact of the land rent differential is manifested in the spatial clustering pattern, whereas large cities without clear economic clusters tend to have lengthier commuting and vice versa (Engelfriet & Koomen, 2017; Song et al., 2017).

Urban size in terms of the population could create apparent commuting pattern differences between high-density urban and less densely sprawling areas (Sultana & Weber, 2007). When a city's population increases, the market responds by containing or reducing people within a location to adjust their interaction costs (Partridge et al., 2010). Higher uncertainty about commuting costs will decrease residential mobility (i.e. permanent relocation of residence) both in the city center and the surrounding area. These commuting costs optimally connect the individual's workplace and dwelling, with a residential relocation postponed to a later point in time. Residents of remote suburban and exurban areas continue to tolerate higher commuting costs (Schmidt, 2014), mainly because the commuting and wage rates relations tend to be convex (Hu et al., 2017).

Commuting between suburbs

While polycentricity emerged in the third wave of urban historical development, commuting between suburbs is not new. It has become prominent in the early twentieth century. During 1912-1936, the annual commuting aggregate between the first-tier suburbs of Chicago exceeded the traditional commuting (i.e. suburb-to-CBD) by at least 3.5 percent (Miller, 2018). From a suburban perspective, out-commuting to other suburbs could be analogous to intercity commuting. This view rules out the hierarchy of urban functions and focuses on the horizontal relationship with different characteristics instead. Suh (1988) has shown that intercity commuting is not possible under the standard urban economic model because an individual who commutes between cities should pay more commuting and rent costs than those who live near the boundary.

Empirical evidence has also emphasized the importance of various settlement and growing employment destination characteristics. Intercity commuting flow supplies are generated by the differences in wages, housing costs, the distance between counties,

and net migration (Renkow & Hoover, 2000). Meanwhile, intercity commuting will increase due to growing job ratios and diversity, and the insignificant accessibility effect suggests competition between destinations (Congdon & Lloyd, 2012). Hence, labor market structure is essential in explaining commuting between municipalities (Watkins, 2016), and the commuting numbers increase in proportion to the city size, population density, and employment (Kofoed, 2020). These studies have shown that under various circumstances, the demand for and supply of employment is an essential determinant of inter-city commuting.

In the case of multi-centered urban areas, the developed subcenter may inherit the CBD characteristics, however, on a smaller scale. Conditional to its attractiveness, the two subcenters would vary in their rental cost and wage levels. The asymmetric land rent between subcenters in a polycentric region leads to incomplete population segregation (Wrede, 2015). Moving costs and market imperfections in housing support the coexistence of people working in different employment centers, explaining the need for cross-commuting (Viguié, 2015). While a concave commuting cost connects between subcenters (Berliant & Tabuchi, 2015), the subsidy for intercity commuting in equilibrium could create welfare-enhancing transfers that stimulate commuting decisions (Borck & Wrede, 2009). Otherwise, cross-commuting workers need to be compensated with higher wages.

Heterogeneity between urban areas, including a life quality differential caused by the discrepancy of endowed amenities, also provides an incentive for intercity commuting. Hypothetically, traveling flows between cities were influenced not only by land rent and wages but also by household structure and their members' skills level (Tscharktschiew & Hirte, 2010). While a household with high-skilled members is assumed to increase commuting distance, having fewer family members increases intra-zonal commuting trips.

Synthesizing inter-suburban commuting: a conceptual framework

As depicted in Figure 1, our conceptual framework suggests that inter-suburban commuting tends to be influenced by unequal land-use distribution between suburbs. Due

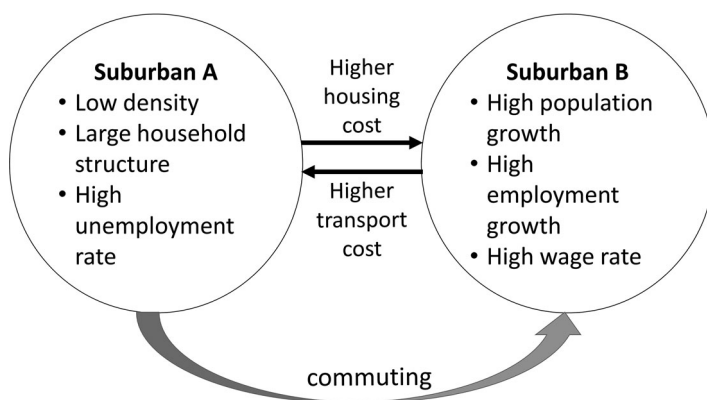


Figure 1. A conceptual framework of inter-suburban commuting.

to the decentralization of employment, several suburbs might shift from primarily dormitory towns into more attractive places and offer greater employment opportunities, better wages, and higher life quality with more access to urban amenities. In contrast, other suburbs in the same metropolitan area merely maintain or decline their residential function. We expect that inter-suburban commuting is a worker's response to match their aspiration with these heterogeneities. For some workers, higher amenity disparities could potentially threaten the utility maximization process and degrade their quality of quality. Therefore, they mitigate this by traveling between different locations.

Suburbs with higher life quality experience population growth, increasing population density, and fewer household members. Suppose locations with increasing population and density do not coincide with the employment opportunities, the unemployment rate will rise to worsen the job-housing balance. The particular employment growth poles can potentially pull workers from other dormitory suburbs or those in decline. However, housing prices close to more attractive suburbs tend to be higher due to land rent competition with business entities. Hence, it attenuates the subcenter housing accessibility, forcing workers to compensate for higher commuting costs with lower housing costs.

Peripheral urban development of Jakarta metropolitan area

Over the last five decades, the development of Jakarta has expanded rapidly to the surrounding areas. Sadewo et al. (2018) have shown that during 1976–2016 the number of medium and large enterprises in the JMA suburbs have grown at an average rate of 93 percent annually. Nevertheless, many service industries in the periphery have lost their significance in the last decade, as most of newly establishments choose to locate in Jakarta. Rukmana et al. (2018) reported that nearly 40 percent of the manufacturing industries in the JMA, including seven large industrial estates, are located in Bekasi Regency. The manufacturing shift from Jakarta to its periphery signifies a regional restructuring process.

There occurred several consequences to the population in the peripheral areas. Jakarta's population growth has decreased from 2.5 percent before the 1990s to 0.11 percent a decade after, followed by an increase in its surrounding areas (Cahyadi & Surtiari, 2016). In the last three decades, a decreasing proportion of Jakarta citizens (from 54.6% to 35.5%) indicates that many people are moving into the periphery (Farda & Lubis, 2018, p. 43). A recent spatial structure of the JMA is polycentric, with at least four significant subcenters located outside Jakarta, including Bekasi, Tangerang, Bogor Regencies, and Bogor (Sadewo et al., 2020).

The spatial restructuring in the JMA has also influenced commuting patterns, including an increase in the average number of daily trips and trip lengths in the JMA between 1985 and 2000 (Susilo et al., 2007). Parallel to the peripheral population growth, the number of daily commuters in the mid-2000s reached 700,000 people, increasing tenfold compared to two decades earlier (Dharmowijoyo et al., 2014). Sadewo (2021) reports that the average commuting length of JMA in 2005 was 21.94 km, with an average travel time of 54.78 min. Most long-distance commuting trips are from outer suburbs like Tangerang Regency (37.03 km). Meanwhile, Bekasi Regency residents have the shortest commuting length of 15.33 km, indicating a difference in employment

attraction for local workers. Although Hudalah et al. (2013) have suggested that the massive employment suburbanization in the JMA took place in the 1990s, the general commuting pattern in the early 2000s is still dominated by the traditional travel to the CBD.

The peripheral urban transformation in the JMA is reflected in the massive growth of urban settlements on the outskirts and the rise of reverse and complex commuting (Firman, 2011). In the early 2000s, almost a million people commuted daily between Jakarta and fringe areas. Numerous workers are employed in peripheral small and new towns while maintaining residence in Jakarta. However, recent evidence shows that the substantial urban spatial transformation in the JMA over the last decade is also followed by increasing inter-suburban commuting. In 2014, there were almost 750,000 commuters between the JMA's peripheral areas, nearly three times bigger than the reverse commuting from Jakarta to surrounding areas (Sadewo & Syabri, 2018). The growing number of commuters increased JMA's average commuting time to 61.87 min in 2015, mainly from inner-suburban commuters (Sadewo, 2021). Rising commuting time indicates a more sprawled JMA development and also manifests widening destination options for the inner-suburban residents compared to a decade ago. The expanding scale of an industrial cluster during 2006–2016 (Sadewo et al., 2018) seems to have contributed to an increase in inter-suburban commuting.

Regarding mobility, social segregation is visible in commuting mode preference. An earlier study by Sugiarto Miwa et al. (2014) in the JMA has shown that increasing income accompanies rising transportation costs. Like many other developing countries, up to 56 percent of commuting travel in the region is conducted by motorcycle (Sadewo & Syabri, 2018). While the automobile-based commuter growth was stagnant between 2000–2010, the use of motorcycles increased by 27.5 percent (Susantono et al., 2011). The use of motorcycle by JMAs' suburbanites can save 30 percent in transportation costs compared to when public transport is used (Rukmana et al., 2018). The rapid use of low-cost private vehicles to facilitate transportation cost-maximization seems to have encouraged many people to keep commuting rather than moving to a different location.

Data and methods

This study takes the suburban perspectives to analyze in- and out-commuting flows. The urban classification in this article is based on the administrative definition. The JMA covered an area of 6,256 km², consisting of eight municipalities and one province. In presidential Decree No. 54/2008, the five municipals within Jakarta are governed as a special region and classified as the urban core of the JMA. In contrast, the surrounding municipals outside Jakarta as JMA's satellite, are traditionally recognized as its suburbs. The JMA suburbs can be grouped into the inner- and outer-peripheries (Kurnia et al., 2022; Rukmana et al., 2018, p. 110). The municipalities adjacent to Jakarta, such as Bekasi, Depok, South Tangerang, and Tangerang, are considered inner-peripheries, whereas Bekasi Regency, Bogor Regency, Bogor City, and Tangerang Regency are classified as outer-periphery (Figure 2). This categorization and their corresponding municipal areas are used as our geographical units of analysis.

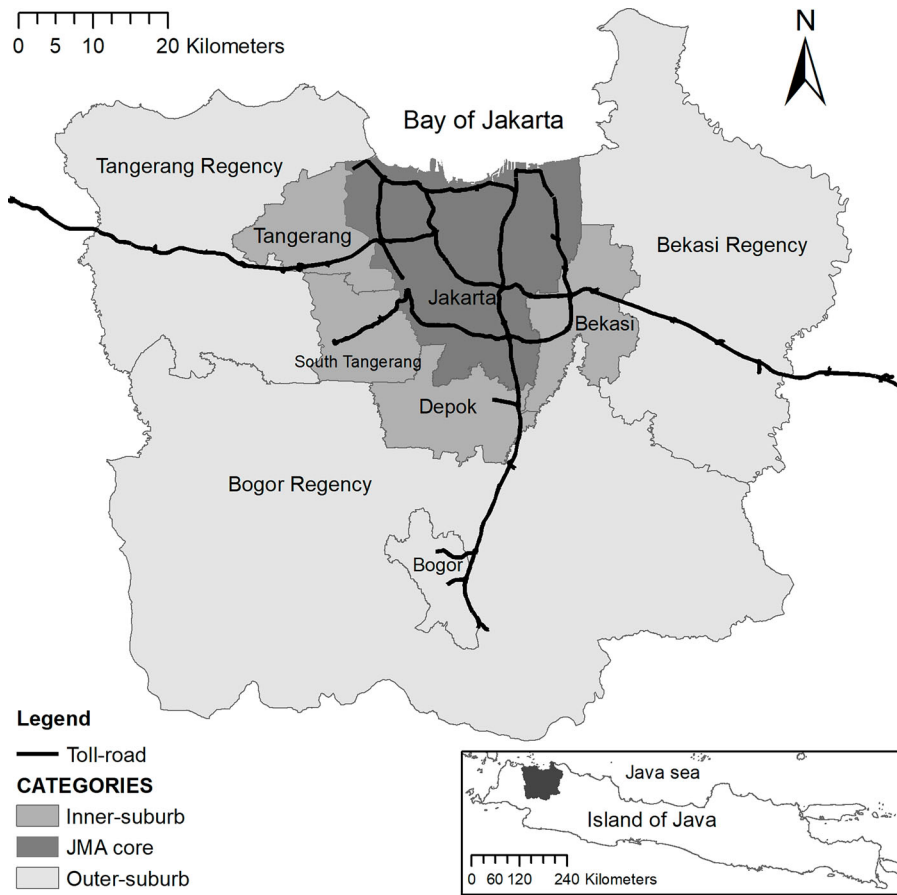


Figure 2. Urban classification of JMA.

This study analyses in-commuting flow (measured as the total number of in-commuters) and out-commuting flow (the total number of out-commuter). It is appropriate to reflect the gap in employment opportunities between counties (Goetz et al., 2010, p. 281). Following the aforementioned conceptual model, several suburbs of JMA generate and attract commuting both from Jakarta and other suburbs due to massive employment growth and change. Following the conceptual framework in the previous section, this study examines the influence of the drivers discussed in the frameworks on in- and out- commuting from the suburbs.

Only a few datasets provide commuting information on a large scale in Indonesia. The national statistical office (BPS) has published commuting data through the annual labor force surveys (SAKERNAS) at the administrative municipality level since 2008. However, the regency-level data is absent from the 2016 SAKERNAS. Therefore, we limited our analysis to 2008–2015 data. Following the SAKERNAS concept, a person is defined as a commuter if they perform daily travel by crossing their administrative residence borders and coming back within 24 h. This study uses various regency-level socioeconomic indicators released by the BPS in the corresponding years as the explanatory variables (see Table 1).

Table 1. Description of variables used in the spatial panel model.

Factors	Factors	Variables / Proxy	Measure	Source
Suburban commuting	In-commuting	Numbers of in-commuters from other suburbs (People)	Ln (in-commuting)	National Labor Survey (SAKERNAS)
	out-commuting	Numbers of out-commuters to other suburbs (People)	Ln (out-commuting)	
	Traditional commuting	Numbers of out-commuters to Jakarta (People)	Ln (out-commuting)	
Suburban heterogeneity	Population growth	Differences in population between particular period	Annual population growth (%)	Municipal in figures publication on respected years
	Density	Density of suburban municipality	Ln (Numbers of people per Km ²)	
	Household (HH) structure	Size of household	Avg. HH members	Economic census 2016
	Wage	Minimum municipal wage (Rupiah)	Ln (wage)	
	Unemployment	Numbers of active workforce which unemployed	Unemployment rate (%)	
	Employment opportunity	Numbers of new employment in manufacture	Ln (new employment)	
	Transport cost	Average per kilometer transportation cost for commuters (Rupiah)	Ln (transport cost)	
	Housing cost	Average housing cost per sq. Meters (Rupiah)	Ln (housing cost)	SAKERNAS & National Socio-Economic Survey (SUSENAS)

The studies of commuting flows at the macro-level have mainly focused on the spatial interaction between geographical regions. Several methods, like gravity and radiation models, might have been widespread. Still, both have been criticized for failing to reflect the influence of socio-geographical segregation and location specialization on commuting flows (Masucci et al., 2013) and overestimating flows (Spadon et al., 2019). Additionally, the gravity model assumes that the residual error of bilateral flows is independent of origin and destination alternatives, which was unlikely to be achieved because the choice of a destination affects the other options (Pace & LeSage, 2008). Ignoring this assumption would not only produce an inefficient estimation but might also yield significant inconsistency and bias. Moreover, there is a need to account for the temporal effects of commuting flow, especially when dealing with urban transformation. While the volume of migration and commuting tends to change over time (Drobne et al., 2013), most studies using spatial interaction models lack temporality as they are conducted by creating a time dummy or repeating the models along the observation period.

To overcome the gravity model drawbacks, a spatial regression model to estimate the bilateral flows type of cases was proposed. An interaction between geographical units is modeled as a spatial lag process, where geographical proximity results in the dependency between origin and destination alternatives. Our paper uses the spatial panel model to estimate the influencing factors of inter-suburban commuting flows in the JMA. There are two advantages of employing this method. First, the enhancement or depression of commuting flows is strongly correlated with the attractiveness propensity of the neighboring destinations (Griffith & Jones, 1980). Therefore, incorporating spatial weights in the model not only accounts for spatial variation but also indirectly considers proximity effects between regencies. Meanwhile, taking a panel structure in the model

provides more variability and degrees of freedom, with less collinearity among variables, making it more efficient and informative (Baltagi, 2008, p. 7). To increase efficiency and mitigate the risk of producing a biased standard error, the spatial panel model accounts not only for serial but also for spatial correlation (Copiello, 2020).

Following Elhorst (2014), the taxonomy of a linear spatial dependence model could be distinguished using two perspectives. First, we can start with a non-spatial regression model and test whether spatial interaction (forward step) is needed. Concerning the panel data structure, the regression model could be written as.

$$Y = \alpha + X\beta + \mu_i + \varepsilon_{it} \quad (1)$$

where Y denotes a vector of $N \times 1$ commuting flow from (to) municipal unit- i in the JMA periphery at time- t ($i = 1, 2, \dots, n$; $t = 1, 2, \dots, T$). Vector Y is paired with $N \times k$ matrix of explanatory variables X . While α is a constant, β is $k \times 1$ vector of unknown coefficients, μ_i is an unobserved individual-specific effect, and $\varepsilon_{it} \sim iid(0, \sigma_\varepsilon^2)$ is a random disturbance. The parameter μ_i can be treated as fixed or random, and the β which is estimated by the random effect ($\hat{\beta}_{RE}$) and the fixed effect ($\hat{\beta}_{FE}$) are assumed to both consistent. Pace and LeSage (2008) described that if $q = \hat{\beta}_{FE} - \hat{\beta}_{RE}$ and $var(q) = var(\hat{\beta}_{FE}) - var(\hat{\beta}_{RE})$ then the Hausman test $m = q'[var(q)]^{-1}q$ has a limited χ^2 distribution with a degree of freedom equal to the number of parameters. If the value of m is statistically significant, the model is favored for a fixed effect (FE) rather than a random effect (RE).

Secondly, one could start with a more general nested model consisting of a spatial interaction effect and then get a more parsimonious model through parameter significance testing (backward step). From this perspective, a complete model with all interaction effects could be written as.

$$Y = \lambda WY + \alpha_N + X\beta + WX\theta + u \quad (2)$$

$$u = \rho Wu + \varepsilon \quad (3)$$

The $n \times n$ nonsingular spatial weight matrix W consist of non-negative pre-specified value on the (i,j) -th element ($i = 1, 2, \dots, m$; $j = 1, 2, \dots, m$) which described the closeness between locations i and j . The spatial interaction parameter λ denotes the endogenous effect, θ denotes the exogenous effect, and ρ denotes the error effect. Combining equations (1) and (2) results in the general nesting model with the spatial panel effect.

$$Y = \lambda WY + \alpha_N + X\beta + WX\theta + u \quad (4)$$

$$u = \mu + \rho Wu + \varepsilon \quad (5)$$

This study uses the latter approach, whereas the parameter estimation and model identification are conducted using the *spml* package (Millo & Piras, 2012) in R software. While applying a maximum likelihood approach for parameter estimation, this package utilizes several additional methods for hypothesis testing, such as the LM test (Baltagi et al., 2003; Baltagi et al., 2007) and the spatial Hausman test (Pace & LeSage, 2008). Before the parameter estimation, error components were examined to concentrate and maximize the model likelihood iteratively. To determine the appropriateness of a spatial panel model, Lagrange Multiplier (LM) test is conducted using the joint BSK method and its extension (Baltagi et al., 2003). Meanwhile, the BSK method is employed

to test for a random regional effect (Baltagi et al., 2007). The conceptual framework for a model selection is given in Figure 3.

This study applies various spatial weights for the model estimation. The models were examined using three different spatial weights and compared against each other (Ganning et al., 2013), including the following: (1) the nearest city weight (W1), (2) inverse-distance weight (W2), and (3) customized weight using relative inter-suburban commuting flows (W3).

$$W1_{ij} = \begin{cases} 1, & \text{if two areas shared similar border} \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

$$W2_{ij} = \frac{1}{d_{ij}} \quad (7)$$

$$W3_{ij} = \frac{\sum_{t=1}^T c_{ij}}{\sum_{t=1}^T c_i} \quad (8)$$

where d_{ij} is the Euclidean distance (in kilometers) between the highest local employment peak in municipals i and j . Meanwhile, c_{ij} denotes a commuting volume (the number of commuters) from a municipal i to a municipal j . All spatial weight values were normalized on rows. The appropriate weight for the model is obtained by comparing an information criterion (Seya et al., 2013). Thus, the best model in this study is selected by comparing the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) values. Traditional commuting flows model estimation is conducted as a comparison to verify the validity of the explanatory variable used in the model.

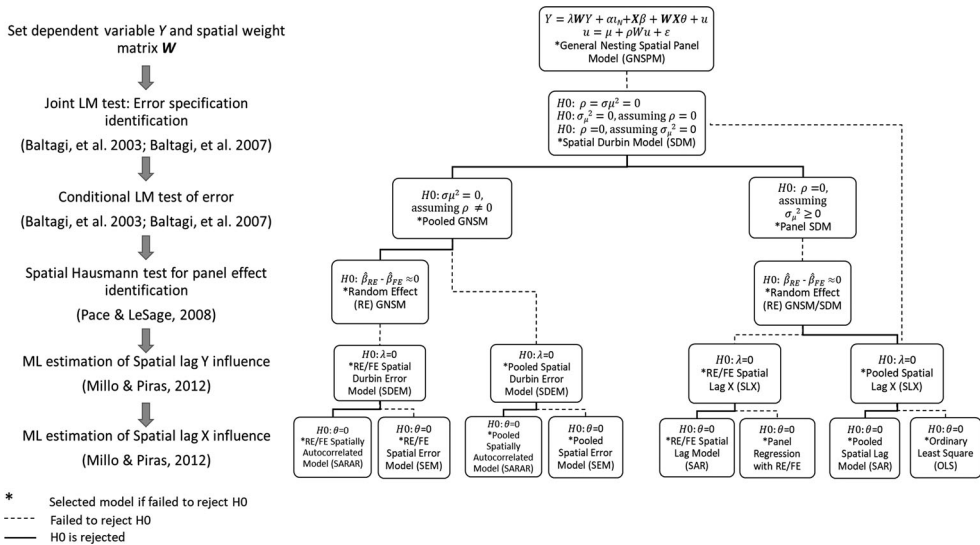


Figure 3. Spatial panel model identification framework.

Results

Growing inter-suburban commuting in JMA

The inter-suburban commuting in the JMA has been increasing during the last decade. Between 2008 and 2015, the inter-suburban annual commuting numbers have grown from 598,000–818,000, rising by 4.65 percent on average. It is higher than the growth of traditional commuting, estimated at 3.7 percent in the same period (Figure 4). Nevertheless, the trips to the CBD still dominate the commuting pattern in the JMA, especially in the eastern region. Bekasi has become the most significant traditional commuter origin in the inner-urban area, while the adjacent Bekasi Regency comes second. Most commuters use private vehicles, growing from 58 percent to 75 percent from 2008–2015. For inter-suburban commuting, Bogor Regency, with its largest population, shared the most considerable portion, followed by Tangerang Regency.

The source of inter-suburban commuting growth not always comes from Bogor Regency but also Tangerang in the inner-urban areas and Bekasi Regency in the outer-suburbs. As in traditional commuting, private modes have become the primary choice for inter-suburban commuters. During 2008–2015, private mode users increased from 48 to 75 percent. The increase of private mode for traditional and inter-suburban commuting mostly came from the mode shift as the public transport use has decreased by half for both the traditional suburb-to-the CBD and inter-suburb commuting flows during 2008–2015.

While there was no specific growth pattern regarding the origin of inter-suburban commuting, the destinations tend to agglomerate in particular locations (Figure 5). On average, Tangerang, Bogor City, and Bekasi Regency account for nearly 52 percent of total in-commuting from other suburban areas. The doubling position of the Bekasi

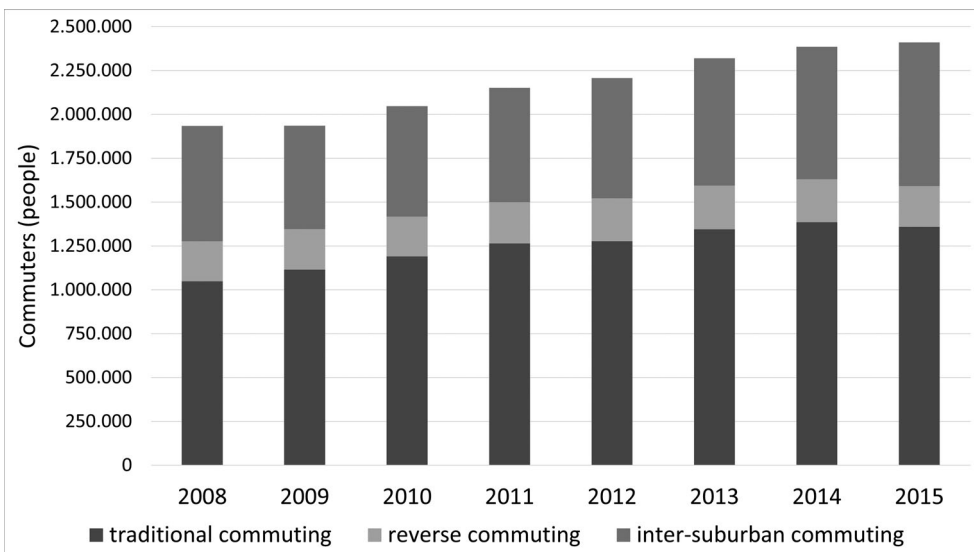


Figure 4. Annual commuting pattern in JMA 2008–2015.

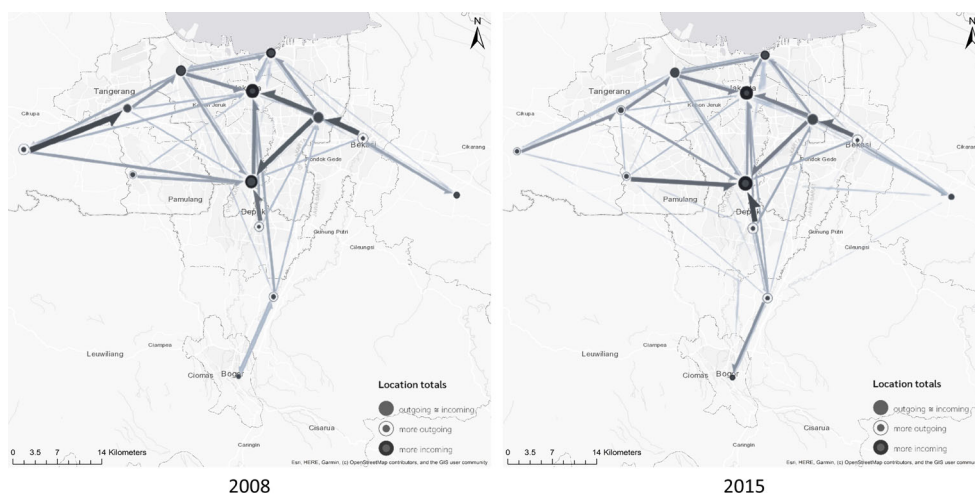


Figure 5. Commuting flow comparison of JMA in 2008 and 2015.

Regency as both a traditional and an inter-suburban commuting origin is somewhat odd, as the area was known as one of the most prominent employment growth poles in the JMA periphery. During 2008–2015, the total employment of the new medium- and large-sized establishment (MLE) in this area reached 126,000 jobs, the largest among other peripheral areas. Most of this employment came from manufacturing industries, shown by an increasing proportion of manufacturing employment (from 30.01 percent in 2008–38.47 percent in 2015) and a rise in the minimum wage by almost three times. In comparison, the Tangerang Regency has generated 19 percent less employment with similar characteristics in the same period, but its traditional commuting was reduced to 12,000 trips.

The massive economic restructuring of manufacturing in the Bekasi Regency has mainly benefitted the workers from the external areas. The job opportunity in Bekasi Regency primarily existed in the manufacturing sectors, while those working in commercial or service sectors had to commute to Bekasi. Burdened by the rapid annual population growth of 4.4 percent between 2008 and 2015, many new jobs in this area do not match local labor qualifications and instead become magnets for workers from other places. For skilled and high-educated service workers, commuting from the Bekasi Regency to Jakarta is still considered beneficial. However, for the marginalized labor, the option was to seek jobs in the nearest suburban neighborhoods.

The adjacent neighborhood, Bekasi City, has the highest average in-commuting growth (Figure 6), implying that this area attracts the most service sector workers from the Bekasi Regency. For the Tangerang Regency, the situation is quite different. Similar economic restructuring waves have brought more benefits to the locals. Having 3.5 percent of annual population growth, the Tangerang Regency had negative in-commuting growth and a low rise in out-commuting. Thus, post-suburbanization in the JMA periphery seems to create a dual impact, benefiting regional and local labor markets in Bekasi and Tangerang Regencies, respectively.

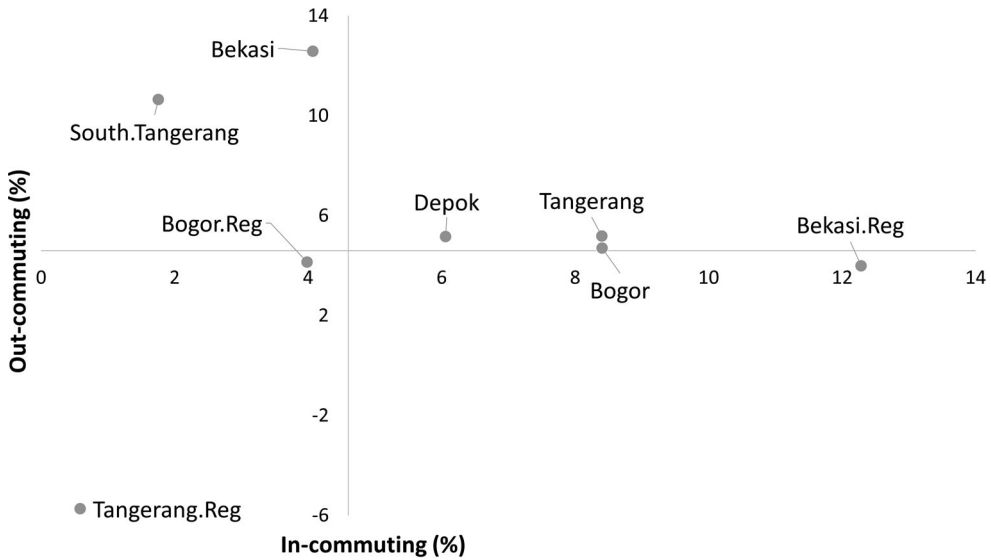


Figure 6. Average growth of JMA's inter-suburban in- and out-commuting between 2008 and 2015.

The suburban commuting models

A spatial panel regression model was applied for both out-commuting and in-commuting flows to verify the influencing drivers of inter-suburban commuting. Prior to the parameter estimation, error components, the appropriateness of a spatial panel model, and random regional effect were examined. The results of the error component model using the BSK method are significant, rejecting the null hypothesis of no spatial error and a random error effect ($H_0: \rho = \sigma_\mu^2 = 0$). The extension test of a random regional effect assuming no spatial error effects ($H_0: \sigma_\mu^2 = 0$, assuming $\rho = 0$) and the BSJK test of a random regional effect hypothesis, which allows for spatial error effects ($H_0: \sigma_\mu^2 = 0$, allowing $\rho \neq 0$), are also rejected. It implies that a random regional effect strongly influences each model, allowing the spatial Hausman test to proceed.

The results of the spatial Hausman test for the different spatial weights are mixed, whereas three models favored a fixed effect (FE) while the rest supported a random effect (RE). However, when comparing its parsimony, the nearest city spatial weight matrix (W1) provides the RE with the lowest AIC and BIC values for each commuting pattern. Thus, the spatial regression parameter estimation proceeds with the RE. The results of the panel effect identification test and a model selection are provided in Table 2.

The parameter estimation results for an inter-suburban commuting model suggested the presence of an interaction effect in each model. While a significant endogenous interaction effect was only found in the in-commuting model, an exogenous interaction effect and interaction among errors could be identified in both in-commuting and out-commuting models (Table 3). Hence, in-commuting flow is estimated under the General Nesting Spatial Model with a random error (GNSM-RE). In contrast, the out-commuting model is estimated under the Spatial Durbin Error Model with a random error (SDEM-RE). The presence of error interaction implies that the model still contains a certain

Table 2. Hypothesis testing result for model identification.

Model identification	out-commuting to suburban			In-commuting from suburban			Traditional commuting		
	W1	W2	W3	W1	W2	W3	W1	W2	W3
$H_0: \rho = \sigma\mu^2 = 0$	10.36**	32.42***	4.76*	20.25***	15.26***	2.47	30.64***	23.68***	21.52***
$H_0: \sigma\mu^2 = 0$, assuming $\rho = 0$	3.22**	5.69***	2.18*	4.50***	3.91***	1.36	5.54***	4.87***	4.64***
$H_0: \sigma\mu^2 = 0$, allowing $\rho \neq 0$	4.15*	3.98*	4.52*	4.31*	2.31	4.32*	3.03	2.54	4.50*
Spatial Hausman test									
$H_0: \hat{\beta}_{RE} - \hat{\beta}_{FE} = 0$	3.09	18.74	86.61***	11.73	31.55*	44.79***	23.01	8.51	4.49
Panel effect	RE	RE	FE	RE	FE	FE	RE	RE	RE
AIC	371.06	428.36	2423.64	348.55	500.27	359.70	351.92	415.70	230.22
BIC	409.92	467.22	2481.39	387.41	558.56	417.99	390.78	454.56	269.08

Table 3. Parameter estimation of the best model selected.

Explanatory variable	Inter-Suburban out-commuting	Inter-suburban in-commuting	Traditional commuting
Intercept	9.39	24.66***	2.09
Pop. Growth	0.00	0.00	0.00
Density	-1.02***	-0.24	-0.38
Household structure	0.1	0.07	0.08
Wage	0.5	-0.75*	-0.01
Unemployment	0.00	-0.01	0.01
Emp. opportunity	0.07**	-0.02	0.02
Transport cost	-0.01	0.04	0.07
Housing cost	0.73*	0.05	1.35***
Sp. lag (Pop. Growth)	0.01	-0.01	0.00
Sp. lag (Density)	-2.03***	-0.77	-1.92***
Sp. lag (Household structure)	0.16	-0.27	0.1
Sp. lag (Wage)	-0.29	1.31***	-0.09
Sp. lag (Unemployment)	0.00	-0.05***	-0.01
Sp. lag (Emp. opportunity)	-0.01	-0.02	0.04
Sp. lag (Transport cost)	-0.02	0.06	0.13
Sp. lag (Housing cost)	2.63**	-0.83	1.94***
ρ (spatial error)	0.55**	0.62***	-0.07
λ . (spatial lag)	-0.49	-0.40*	-0.13

amount of an effect due to the omitted variables that are spatially autocorrelated between geographical units.

For the out-commuting model, an increase in population density by one percent would decrease out-commuting to another suburb by an equal magnitude. Conversely, an increase in housing costs by one percent could improve out-commuting by 0.7 percent. For both variables, there is a significant exogenous interaction with a similar effect. The exogenous interaction of the SDEM-RE model could be directly interpreted as the indirect or spillover effect resulting from the change in explanatory variables (LeSage & Pace, 2009, p. 42). Thus, an increase in population density in a particular suburban area by one percent would also decrease not only out-commuting activities in that suburb but also in the surrounding areas by two percent. Similarly, an increase in housing costs by one percent would improve out-commuting flow by 2.6 percent. A significant positive influence can also be found in the new employment opportunities in manufacturing with a smaller magnitude. This suggests that people who commute to other suburban areas seek employment opportunities other than manufacturing.

In the JMA periphery, high residential density with more expensive housing costs characterizes the inner-suburban area. The model's contrasting effects on housing costs are visible through the high out-commuting flows from this area. An agreement with the theory is reached if the impact of both density and housing costs are negative (i.e. commuters come from more minor dense locations with lower housing costs). This means that only the housing cost variable contrasts with our proposed conceptual framework. During 2008–2015, the average number of inner-to-outer suburban commuting reached 42 percent, while the reverse direction accounted for 28 percent (Figure 7). For instance, a lower population density with a high out-commuting number could be found in Bogor Regency and Tangerang Regency. Many workers from those areas travel to their adjacent neighborhoods, Bogor and Tangerang. However, it should also be noted that during 2008–2015 both h experienced vast growth in out-commuting to other suburbs. Thus, there is an out-commuting exchange between paired

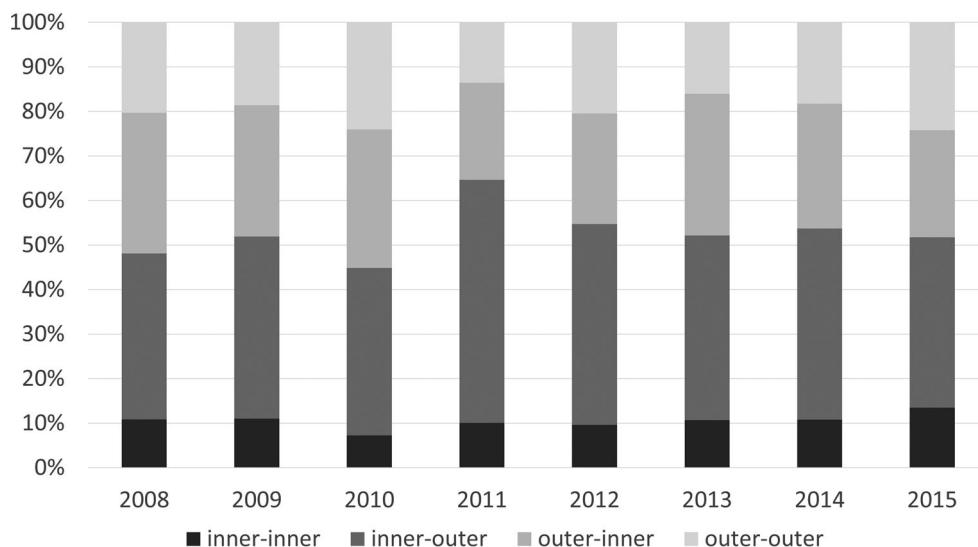


Figure 7. The dynamics of annual inter-suburban commuting direction in JMA 2008–2015.

inter-suburban areas. The workers commute from the metropolitan edge to the inner-suburbs. In return, a significant number of those workers made the reverse commuting from the inner-suburbs to the edge cities.

The negative endogenous interaction effect confirms the commuting exchange between paired suburban conjunctures. It implies that the number of in-commuting in a location would only increase if in-commuting in the neighboring areas decreased. This strengthens previous findings that the inter-suburban commuting destination is centralized only in several places. The mechanism is explained by the increasing in-commuting numbers in the areas surrounded by municipalities with lower unemployment rates. Another exogenous interaction for in-commuting exists with wage levels and unemployment rates. While higher wages are strongly associated with the existence of a manufacturing subcenter in the outer-suburb, commuting for non-manufacturing employment has created a negative direct effect on wages. On the contrary, the reverse commuting to a lower-wage area provides a positive indirect impact on the surrounding areas. Hence, the manufacturing-led post-suburbanization process in the JMA seems to have benefited only certain suburban groups.

The driving forces of inter-suburban commuting in the JMA tend to be similar in many ways to those of traditional commuting. From a suburban viewpoint, the panel specification of the best traditional commuting model was the RE with W3. The traditional commuting in the JMA is strongly affected by housing costs, either in the suburbs themselves or in their neighboring areas. A positive housing cost coefficient and a negative spatial lag of a distance suggested that traditional commuting decays with the growing distance from the CBD, confirming the monocentric urban model. Municipalities adjacent to Jakarta, such as Bekasi, Tangerang, Depok, and South Tangerang, have become prominent residential locations for traditional commuters. Those areas have more expensive housing costs due to their proximity to Jakarta, which also made the effect of neighboring inner-suburbs with similar high housing costs positive. However, the

model does not show a significant spatial lag and spatial error, suggesting that both endogenous and error interaction do not influence traditional commuting. In other words, the dynamic of density and housing costs are sufficient to explain the traditional commuting pattern in the JMA.

Discussion

From a spatial point of view, the inter-suburban commuting in the JMA exogenously interacts with neighboring urban heterogeneity. Commuting between suburban areas is strongly affected by the changing urban environment (density, housing cost, wage level, and unemployment rate) in its surrounding areas, and neighboring suburbs have become their complementary. Main inter-suburban commuting flows occur laterally between adjacent municipalities along the toll road corridors due to the increased private vehicle usage (up to 26 percent during 2008-2015). More out-commuting occurs from the outer-suburbs of the JMA to the inner-suburbs, showing that the inner-suburbs have achieved a different attractiveness than the CBD and outer-suburbs. It also indicates that the commuting pattern corresponds to employment concentration along the corridor (Lowe, 1998). The sizeable reverse commuting numbers from the inner- to the outer-suburbs were also considerably high, explaining the positive influence of the housing cost factors on the out-commuting model.

The effect of new employment opportunities and wages on inter-suburban commuting is opposite to what has been found elsewhere, differentiating the JMA from other countries. Significant manufacturing employment emerged in the eastern and the western parts of the JMA's outer-suburbs (Hudalah et al., 2013). In line with Timothy and Wheaton (2001), outer-suburbs have also been the fastest growth- and highest wage zones across the JMA. However, several problems arose due to the services and commercial activities' re-concentration in Jakarta between 2006 and 2016 (Sadewo et al., 2018). First, many non-manufacturing workers seek employment elsewhere because it is rarely found in the Bekasi and the Tangerang Regencies. While conforming to the role of industrial land-use in promoting out-commuting (Liu et al., 2020), it may also disprove the reduced significance of residential location importance (Elldér, 2014).

Secondly, higher wages seem to have attracted commuters from inner-suburbs but did not reduce out-commuting from the outer-suburbs. The negative influence of wages on commuting is inconsistent with some previous studies (e.g. Renkow & Hoover, 2000; Suh, 1988; Tscharaktschiew & Hirte, 2010). The anomaly could be explained by the JMAs' commuters characterized by high education with high-end commercial and service employment preferences (Raijaya & Chotib, 2020). Therefore, they have to seek employment in other suburbs. The case of the JMA shows that a high suburban wage rate does not always benefit all residents because Sadewo et al. (2018) finding show that services firms tend to re-concentrate in the core city. Thus, significant travel activities to the core are inevitable for accessing specific service needs.

A strong interdependency between the adjacent suburban job locations is reflected in the spillover effect of independent variables. Density and housing costs influencing out-commuting flows are almost twice as much as the direct effects (feedback). The negative spillover effect of density shows that the high population density which characterizes the inner-suburban areas would lower out-commuting flows in the neighboring inner-

suburbs. For the housing costs, the opposite effect occurred, making only inner-suburbs with the highest density, like Bekasi and Tangerang, which are close to the expensive residential areas like Depok and South Tangerang, capable of sending out commuters in a considerable number. Meanwhile, the higher wages in the post-suburban areas may negatively influence in-commuting flows. Consequently, the spillover goes to the neighboring areas and creates a reverse impact. As urban development in the JMA spreads along the toll road corridors (Firman, 2011), the municipalities passed by these corridors would also benefit from increased employment opportunities. However, this could also bring potential problems in the future because inter-suburban commuters share the same corridors with the traditional and reverse commuters.

Inter-suburban commuting in a particular suburb can be determined by the corresponding land-use pattern in its surrounding areas. For instance, the land-use mix of industrial and commercial centers potentially attracts more in-commuting (Liu et al., 2020). However, manufacturing post-suburbs like Bekasi and Tangerang Regencies have higher minimum regional wage rates than Jakarta. For service firms, this situation diminishes the potential for lower production costs if located near the manufacturing post-suburbs. Thus, it prevents manufacturing and service firms from agglomerating in JMA's post-suburbs. The service firms are more likely to be located in the inner-suburbs because although it has a slightly lower level of wages compared to the post-suburbia, the inner-suburban areas may still provide more amenities due to their proximity to Jakarta.

Conclusion

This article explores how the underlying urban form factors influence inter-suburban commuting in the Jakarta Metropolitan Area (JMA). The factors analyzed in this article represent the unequal distribution of JMA peripheral characteristics, including population growth and density, household structure, employment opportunity, unemployment rate, wages, housing costs, and transportation costs. This article applies the spatial panel regression approach to examine the spatio-temporal effects of these factors on inter-suburban commuting activity in the JMA between 2008 and 2015.

This paper contributes to a better understanding of intra-metropolitan mobility by providing empirical evidence of the effect of employment decentralization and changing suburban environments on inter-suburban commuting patterns. The post-suburbanization process is expected to mediate the transformation of suburbs from metropolitan hinterlands into multifunctional cities (Phelps & Wood, 2011). However, the case of the JMA indicates that manufacturing-led post-suburbanization has a contrasting impact. While it does not always reduce out-commuting, it forces certain groups of localities to seek jobs elsewhere, potentially leading to a worsening job-housing balance in JMA and other metropolises with a similar trajectory of post-suburbanization.

Several policy recommendations can be put forward in line with the analysis results. First, the effectiveness of wage policy as a commuting attractor should be accompanied by employment diversity policy incentives. To better accommodate the impact of the amplifying manufacturing employment clusters phenomenon on urban land-use change, industrial park development needs to be integrated into land-use planning policy. Land-use integration shall be coupled with decentralized transport infrastructure

planning at the local level to improve the connection between inner- and outer-suburbs. A more robust suburban transport infrastructure can anticipate the incoming inter-suburban commuting growth due to increased horizontal mobility between outer-suburbs. Furthermore, an integrated transit-oriented development surrounding the JMA suburbs should be considered a critical urban development strategy to reduce car dependence while at the same time supporting inter-suburban mobility growth in the future.

Irrespective of the finding, this article is subject to several limitations. First, municipal scale administrative delineation for JMAs' urban classification may not describe suburban land-use effectively. Following Sadewo et al. (2020), suburban employment centers cover only a limited suburban area. Future studies should explore this method at a smaller spatial scale to produce more accurate land-use and spatial interaction between suburbs. Another limitation is the use of commuting flows in the model. Most commuting studies use time or length variables as dependent variables to provide a more precise illustration of spatial friction. Therefore, future suburban commuting studies should try to have a specific time and length differentiation between traditional and inter-suburban commuting and compare it to their models.

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